

Nº de funções por SO

	Contiki 3.0	FreeRTOS 10.1.0	Mbed-OS 5.9.6	RIOT 2018.07	TinyOS 1.1.0	TizenRT 1.1	Zephyr 1.12.0	Total
_snprintf	1					3		4
access	3	2	44	2		16	11	78
atoi	25		13	204	22	471	14	749
atol	1		2	9	2	7		21
bcopy						2		2
char	1042	1	1670	777	72	3225	1106	7893
chgrp						2		2
chmod				2		3		5
chown				2		1		3
CopyMemory	4							4
CreateProcess	2							2
crypt			2			116		118
drand48						5		5
EnterCriticalSection						2		2
equal		40	9			3	3	55
EVP_des_cbc						1		1
EVP_rc2_40_cbc						1		1
execl				2		2		4
execle				2				2
execlp	1			2				3
execv				2		7		9
execvp				2		1		3
fgetc	5		7			18		30
fopen	10		84	3	8	243	14	362
fprintf	2		55	1		59	45	162
fscanf	3		3					6
fwprintf						1		1
fwscanf						1		1
getc	8		35		26	14		83
getchar	1		5	7		82	1	96
getenv	1			1	9	44	1	56
getlogin				2				2
getopt	10			5	3	61		79
getopt_long	3			1				4
gets			4			5		9
getwd						7		7
InitializeCriticalSection						2		2
LoadLibrary	5					3		8
memalign				1		5		6
memcpy	574	9	2062	557	34	1859	1194	6289
mismatch			70					70
mkstemp						6		6
mktemp						4		4
MultiByteToWideChar			1			6	1	8
open	44		153	20	8	245	28	498
popen						1		1
printf	210	2	175	17	5	417	68	894

Nº de funções por SO

random			40	39		73	27	179
read	52	5	823	91	8	275	139	1393
readlink				2		8		10
realpath						13		13
scanf	6		6		4	21		37
snprintf	27		19	2		97	23	168
sprintf	78	5	93	14	4	104	51	349
srand	4		50			33	1	88
srandom						2	1	3
sscanf	2		15			20		37
strcat	16		17	2	18	56	4	113
strcpy	104	1	63	18	25	102	42	355
strlen	194	4	606	203	21	2222	372	3622
strncat	5		7	7		62	2	83
strncpy	125		55	21	3	333	36	573
swprintf						1		1
syslog						16		16
system	15	66	1	7	2	14	5	110
TCHAR			4	1		3	8	16
tempnam						2		2
tmpfile	1		1			1		3
tmpnam			1			2		3
umask				2				2
usleep	34		4	12	4	179	20	253
vfork						7		7
vfprintf	8		4	5	3	32	6	58
vfprintf	1		2					3
vfwprintf						1		1
vprintf	1		8	1		12	3	25
vscanf			7					7
vsprintf	13		8	4	2	52	8	87
vsprintf	1		7			3	3	14
vsscanf			5			5		10
vswprintf						1		1
vwprintf						1		1
wcscat						1		1
wcscpy						1		1
wcslen						6		6
wcsncat						1		1
wcsncpy						1		1
wprintf						1		1
wscanf						1		1
TOTAL	2642	135	6240	2052	283	10716	3237	

Nº de linhas de código

	Contiki 3.0	FreeRTOS 10.1.0	Mbed-OS 5.9.6	RIOT 2018.07	TinyOS 1.1.0	TizenRT 1.1	Zephyr 1.12.0	Total
Nº de linhas	240691	67292	3841499	1191661	12536	1055560	3040075	9449314
Nº de arquivos	556	50	1078	480	43	1538	580	4325
Média Vuln./Arquivo	4.75	2.74	5.80	4.28	6.65	7.06	5.59	
Função insegura	2642	135	6240	2052	283	10716	3237	25305
Porcentagem func. Inseg / linha de cod.	1.09767	0.2006182	0.1624366	0.1721966	2.2575	1.0151957	0.106478	

Severidade

	Contiki 3.0	FreeRTOS 10.1.0	Mbed-OS 5.9.6	RIOT 2018.07	TinyOS 1.1.0	TizenRT 1.1	Zephyr 1.12.0	Total
1	424	49	1627	344	72	3219	577	6312
2	1779	15	4075	1589	155	6177	2430	16220
3	25	0	91	46	12	249	30	453
4	392	71	442	67	44	1040	200	2256
5	2	0	5	6	0	31	0	44

CWE

	Contiki 3.0	FreeRTOS 10.1.0	Mbed-OS 5.9.6	RIOT 2018.07	TinyOS 1.1.0	TizenRT 1.1	Zephyr 1.12.0	Total
CWE-119!/CWE-120	1042	1	1674	778	72	3228	1114	7909
CWE-120	900	15	2290	618	84	2538	1327	7772
CWE-120, CWE-20	91	5	908	104	41	489	140	1778
CWE-120/CWE-785!						13		13
CWE-126	194	44	685	203	21	2231	375	3753
CWE-134	269	2	288	31	10	708	159	1467
CWE-190	26		15	213	72	478	14	818
CWE-327	4		92	39		231	29	395
CWE-362	54		237	27	16	501	42	877
CWE-362, CWE-20				2		8		10
CWE-362/CWE-367!	3	2	44	2		16	11	78
CWE-377	1		2			15		18
CWE-676	34		4	13	4	184	20	259
CWE-732				2				2
CWE-78	18	66	1	17	2	25	5	134
CWE-807				2				2
CWE-807, CWE-20	1			1	9	44	1	56
CWE-829, CWE-20	5					3		8
Total	2642	135	6240	2052	331	10712	3237	

Nº de funções por SO compacta

	Contiki 3.0	FreeRTOS 10.1.0	Mbed-OS 5.9.6	RIOT 2018.07	TinyOS 1.1.0	TizenRT 1.1	Zephyr 1.12.0	Total	Sever.
char	1042	1	1670	777	72	3225	1106	7893	2
memcpy	574	9	2062	557	34	1859	1194	6289	2
printf	210	2	175	17	5	417	68	894	4
read	52	5	823	91	8	275	139	1393	1
sprintf	78	5	93	14	4	104	51	349	4
strcpy	104	1	63	18	25	102	42	355	4
strlen	194	4	606	203	21	2222	372	3622	1
system	15	66	1	7	2	14	5	110	4
Others	373	42	747	368	112	2498	260		
TOTAL	2642	135	6240	2052	283	10716	3237		

CWE compacta

	Contiki 3.0	FreeRTOS 10.1.0	Mbed-OS 5.9.6	RIOT 2018.07	TinyOS 1.1.0	TizenRT 1.1	Zephyr 1.12.0	Total
CWE-119!/CWE-120	1042	1	1674	778	72	3228	1114	7909
CWE-120	900	15	2290	618	84	2538	1327	7772
CWE-120, CWE-20	91	5	908	104	41	489	140	1778
CWE-126	194	44	685	203	21	2231	375	3753
CWE-134	269	2	288	31	10	708	159	1467
CWE-78	18	66	1	17	2	25	5	134
Others	128	2	394	301	53	1497	117	
Total	2642	135	6240	2052	283	10716	3237	